



Dr. Rushikesh Amrut

Urban Data Scientist & Mobility Researcher

 Personal Webpage

 Hamburg, Germany

 LinkedIn Profile

 Google Scholar Profile

 English [C2], German [B2]

Education

Ph.D. - Transportation Systems Engineering

2013 - 2021

Indian Institute of Technology Madras, India

M.Tech. - Transportation Engineering & Planning

2011 - 2013

NIT Surat, India

BE - Civil Engineering

2005 - 2009

Shivaji University, India

Profile Statement

Trained urban data scientist and mobility researcher with 10 years of experience working at the intersection of mobility, geospatial analytics and urban systems in international contexts. I develop data-driven tools, digital workflows and decision-support frameworks that translate complex urban data into actionable insights for policy, planning and implementation.

Skills

- **Urban data & Geospatial:**
QGIS, OpenStreetMap, OSMnx, GeoPandas
- **Modeling & Digital methods:**
Python, R, SQL, MATLAB, VISSIM, SPSS
- **Research & Teaching:**
International Publications, Research proposals, Conference Presentations, Organization of Workshops & Conferences, $\LaTeX$,

Relevant Work Experience

Lecturer

01/2025 - Present

International School of Management, Hamburg

- Designed and delivered 5 university courses (120 hrs) in Statistics, AI, Machine Learning and Data Interpretation for post graduate students with mixed technical backgrounds.
- Developed hands-on learning material and practical exercises that translate quantitative methods into usable analytical and decision-support skills.
- Received 4.8/5 average student feedback, demonstrating strong teaching, capacity-building and technical communication skills.

Postdoctoral Research Associate

08/2022 - 08/2024

Chair of Transport Modeling and Simulation
Technische Universität Dresden, Germany

Focus: Automated planning model for benchmarking road-traffic carbon emissions in 45 global cities 

- Built scalable geospatial data pipelines processing $10M^+$ datapoints.
- Automated Python & R workflows, improving analytical efficiency by 40%.
- Developed and bench-marked decision-support framework that integrates network, travel demand, & carbon emissions for cities worldwide.
- Developed policy level insights for carbon-reduction strategies (SDG 11.6), travel demand management, and investments towards sustainable mobility initiatives.

Impact : ► Simulated scenario-based emissions database of 45 global cities; ► Published the framework in *Discover Cities*, Springer Nature, 2025.

Personal Transition

04/2021 - 06/2022

Family caregiving during COVID-19, relocation to Germany and intensive German language learning.

Achievements

- > Published 15 peer-reviewed journal articles, book chapters and international conference papers.
- > Best paper awards at international conferences:
 - UMI-CODATU 2017
 - ACODS 2020
- > Contributor to the first Indian Highway Capacity Manual (Indo-HCM).
- > Government of India scholarships for:
 - Ph.D. program $\approx$ €16K (<5% acceptance rate)
 - Master's Program $\approx$ €2K (<10% acceptance rate)
 - Travel grant $\approx$ €1.6K to represent the Institute internationally.

Strengths

- > Spatial data & mobility systems
- > Interdisciplinary research
- > Teaching & capacity building
- > International collaboration

Doctoral Researcher

07/2013 - 03/2021

Department of Civil Engineering,
Indian Institute of Technology Madras, India

Focus:

Develop statistical, mathematical and machine-learning models for complex mixed-traffic behavior in Indian cities.

- Collected, curated and analyzed large-scale traffic data from video, fixed sensors and mobile sensors in heterogeneous urban traffic conditions.
- Developed *predictive models for motorcyclist maneuvering decisions*, comparing statistical and ML methods including Random Forest, XGBoost and ANN. [↗](#)
- Co-developed image-processing software that converted raw traffic video into structured trajectory data and improved processing efficiency by 500%. [↗](#)
- Designed automated data-processing, visualization and model-building pipelines in MATLAB and R and guided collaborative master's thesis work.

Impact : ▶ Produced methods relevant to traffic safety, simulation and urban mobility planning in Global South contexts; ▶ Published 10⁺ research articles in leading journals and conferences.

Visiting Researcher

06/2012 - 06/2013

Central Road Research Institute of India
(CRRI), New Delhi, India

- Performed statistical analysis and predictive modeling using SPSS and MATLAB on urban traffic datasets.
- Developed field data-collection protocols and managed 10 master's students across surveys in 5 Indian cities.

Impact: ▶ Official contributor to the development of the first Indian Highway Capacity Manual (Indo-HCM).

Coordination & Knowledge Transfer

- **International research coordination:** Coordinated inputs for 2 grant proposals and 4 publications at TU Dresden, linking mobility modeling, spatial data science and policy-facing urban research.
- **Industry collaboration & applied research:** Collaborated on an applied research project with Tata Consultancy Services (TCS), Chennai Municipal Corporation, and State Transport Services, contributing to the development of data-driven tools for urban mobility analysis, to align technical development with real-world implementation needs.
- **Teaching and capacity building:** Designed applied teaching in statistics, AI, machine learning, SQL and data interpretation for students with varied technical backgrounds.
- **Multi-institutional collaboration:** Worked with CRRI, NIT Surat, IIT-linked partners and city-level survey teams on national transport research supporting the first Indian Highway Capacity Manual.

Reviewing

Journals:

- Computer-Aided Civil and Infrastructure Engineering, Wiley Publications
- Transportation Letters, Taylor & Francis
- Engineering Elsevier
- Journal of The Institution of Engineers (India): Series A Springer

Conferences:

- Annual Meetings of Transportation Research Board, USA

Referees

- ✉ **Prof. S. Travis Waller**
Lighthouse Professor,
Chair of Transport Modeling and Simulation,
TU Dresden, Germany.
✉ steven_travis.waller@tu-dresden.de
- ✉ **Dr. Sai Chand G.**
Assistant Professor,
Department of Civil Engineering,
IIT Delhi, India.
✉ saichand@iitd.ac.in
- ✉ **Prof. Gaurang J. Joshi,**
Head,
Department of Civil Engineering,
NIT Surat, India.
✉ gj@ced.svnit.ac.in
- ✉ **Dr. Shriniwas S. Arkatkar,**
Professor,
Department of Civil Engineering,
NIT Surat, India;
Email: sarkatkar@gmail.com

Developed Projects & Digital Tools

- **Urban emissions decision-support model:** Built a scalable emissions-modeling framework combining traffic, geospatial and validation data to benchmark road-traffic carbon emissions across 45 global cities. *Discover Cities*, Springer Nature, 2025 ✉
- **Staggered-following behavior analytics:** Built an R/MATLAB workflow using copula methods to model the joint lateral and longitudinal following behavior of motorcyclists in mixed traffic. *Traffic and Granular Flow 2022*, Springer, 2024 ✉
- **Powered two-wheeler maneuver prediction:** Developed ML and statistical models for lateral movement decisions using engineered trajectory features, Random Forest, XGBoost, ANN and SVM. *Transportation Letters*, Taylor & Francis, 2022 ✉
- **Mixed-traffic trajectory extraction pipeline:** Co-developed VeTre, a semi-automated computer-vision workflow converting raw traffic video into validated 5M⁺ microscopic trajectory datapoints. *Transportation Letters*, Taylor & Francis, 2021 ✉
- **Mixed-traffic path generation and tracking:** Guided a MATLAB/IPG CarMaker prototype for leader selection, overtaking feasibility, trajectory generation and vehicle path tracking in disorderly traffic. *IFAC-PapersOnLine*, 2020 ✉
- **Real-time traffic-density estimation:** Implemented and evaluated nonlinear filtering approaches, including the Unscented Kalman Filter, for urban mixed-traffic density inference. IEEE ITSC, 2015 ✉

Selected Publications

Journal Articles:

- Waller S.T., **Amrutsamanvar R.**, Qurashi M., Sai Chand G., and Polydoropoulou A. (2025) Automated Planning Model for Estimating and Benchmarking Road Traffic Carbon Emissions in Global Cities. *Discover Cities*, Springer Nature. ✉
- **Amrutsamanvar R.** (2022) Modeling Lateral Movement Decisions of Powered Two-Wheelers in Disordered Heterogeneous Traffic Conditions, *Transportation Letters*, Taylor & Francis, 14 (3), 195-214. ✉
- **Amrutsamanvar R.**, Muthurajan B., and Vanajakshi L. (2021) Extraction and Analysis of Microscopic Traffic Data in Disordered Heterogeneous Traffic Conditions, *Transportation Letters*, Taylor & Francis, 13 (1), 1-20. ✉
- Deshpande P., **Amrutsamanvar R.**, and Subramanian S. (2021) Vehicle Path Generation and Tracking in Mixed Road Traffic. *International Federation of Automatic Control (IFAC)-PapersOnLine*, 53(1), 524-529. ✉
- **Amrutsamanvar, R.**, and Vanajakshi, L. (2024) Two-Dimensional Following Behavior Analysis of Powered Two-Wheelers Using Copula Approach. In: Rao, K.R., Seyfried, A., Schadschneider, A. (eds) *Traffic and Granular Flow 2022*. Lecture Notes in Civil Engineering, vol 443. Springer, Singapore. ✉

International Conference Papers:

- **Amrutsamanvar R.**, Sai Chand G., Qurashi M., and Waller S.T. (2023) Rapid Planning: Opportunities with Pervasive Data for Sustainable Mobility, *Smart Cities Symposium (1-6)*, IEEE. ✉
- Dhivyabharathi, B., Fulari, S., **Amrutsamanvar, R.**, Vanajakshi, L., Subramanian, S., and Panda, M. (September, 2015). Performance Comparison of Filtering Techniques for Real Time Traffic. Density Estimation under Indian Urban Traffic Scenario. In *proceedings 18th International Conference on Intelligent Transportation Systems Conference (ITSC)*, IEEE, (pp. 1442-1447). ✉